

CFE230

Basic computer knowledge and computer evolution



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Keywords:

Computer evolution; basic computer knowledge; Mathematical calculation in computer

Objectives

- To tell the computer development from mechanical calculator to the computer system.
- to tell difference between microprocess and microcontroller.
- To express electronic devices and electronic measurement tools
- Be able to describe the concept of data transferring and detection errors in the computer system.
- Be able to tell mechanical protection and safety in the computer system and IoT

Topic

- From Mechanical machine to Computer
 - Mechanical
 - Vacuum tube
 - Silicon Transistor
- Microprocessor, microcontroller, DSP, GPU
- Quiz (5 points)

1.1.1 Mechanical computer

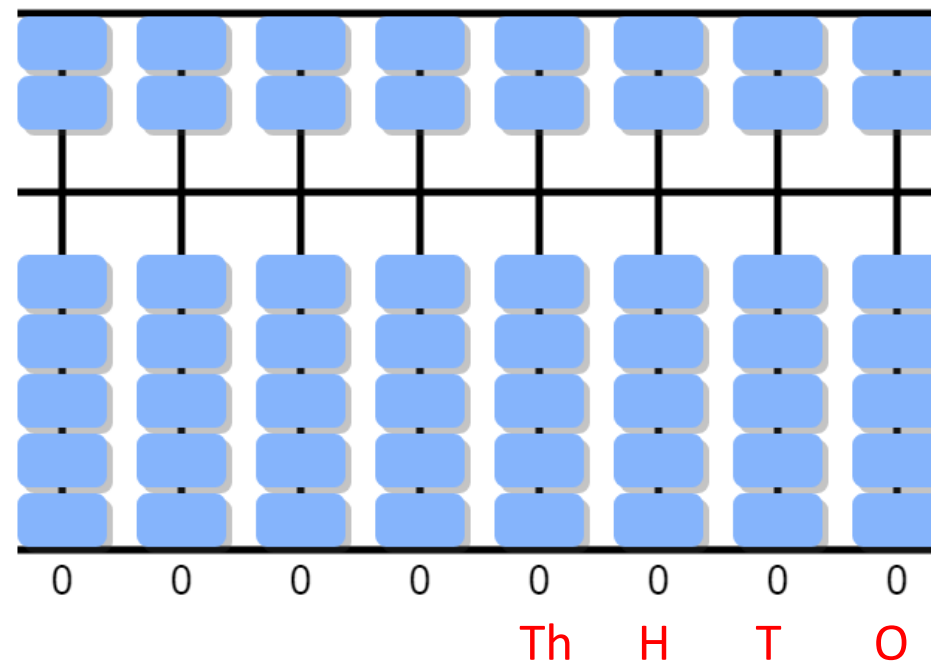
- Abacus
 - Used in China, Europa and Russia



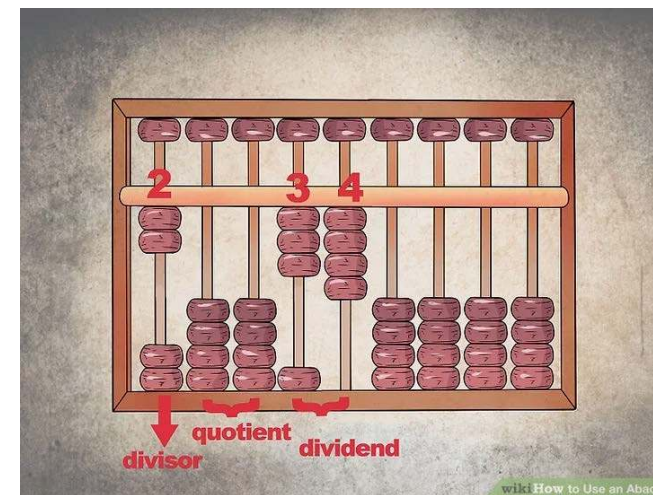
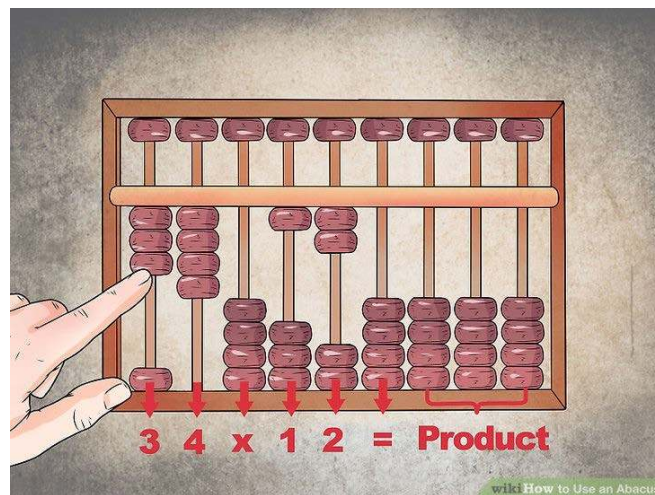
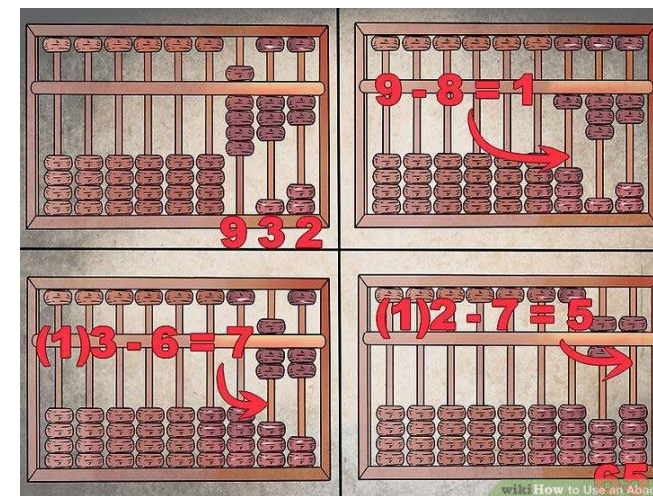
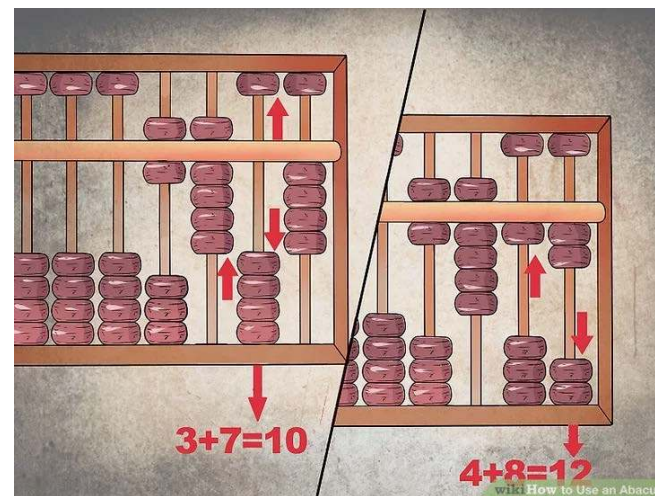
Activity 1.1 Play abacus

- Click on <https://www.mathematik.uni-marburg.de/~thormae/lectures/ti1/code/abacus/sanpan.html>

- Calculate
 - $83 + 12 = ?$
 - $112 + 12 = ?$



More functions in Abacus



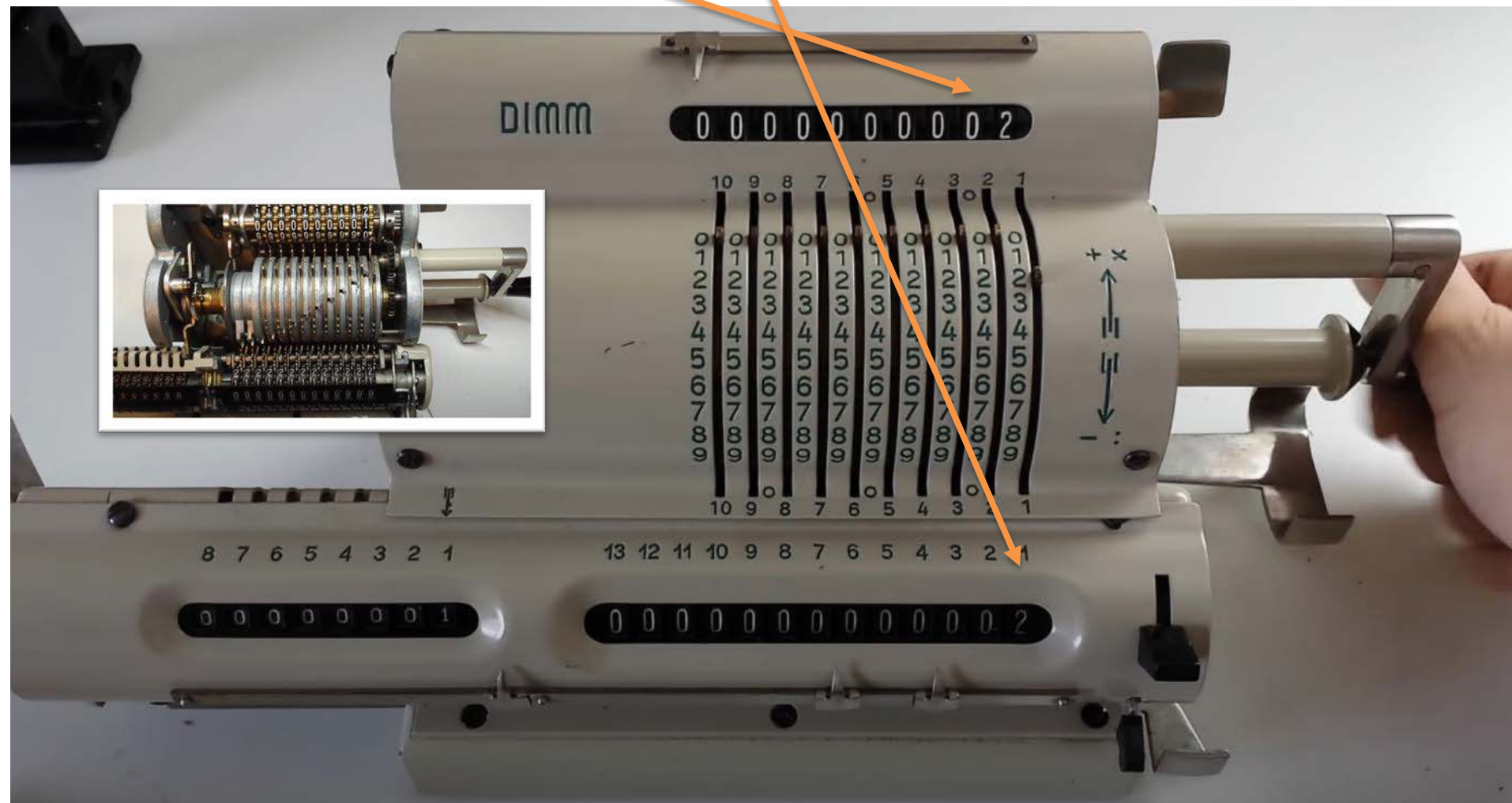
<https://www.wikihow.com/Use-an-Abacus>

1.1.1 Mechanical computer



- Mechanical calculator
 - Blaise Pascal (1642) developed a machine to help his father working in the shop.
 - Working only function of the addition and subtraction.

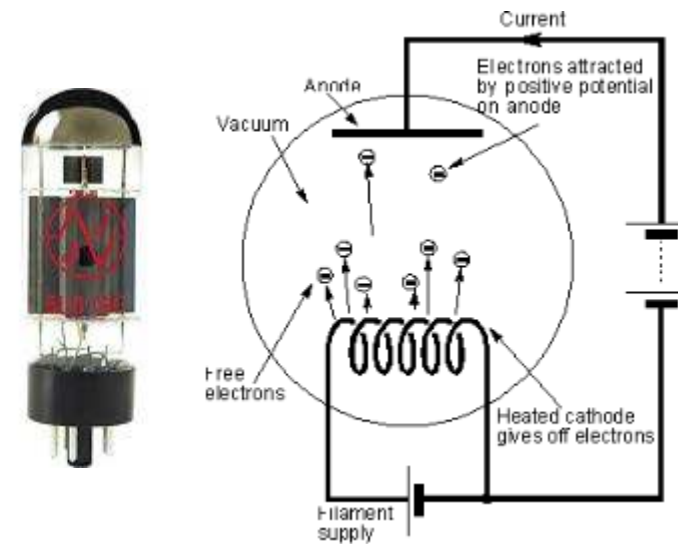
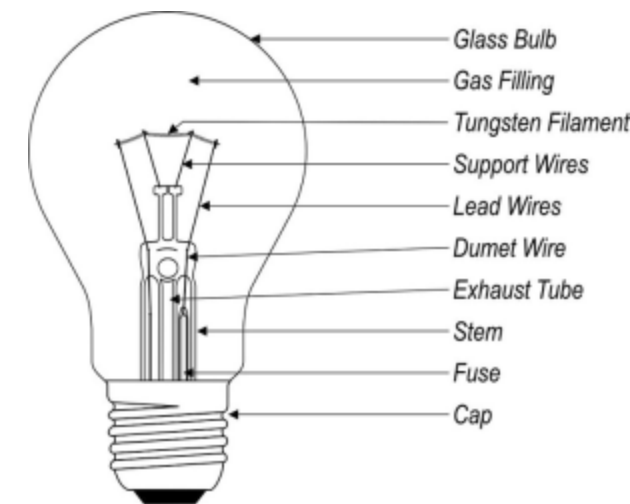
$$2+2 = 4$$

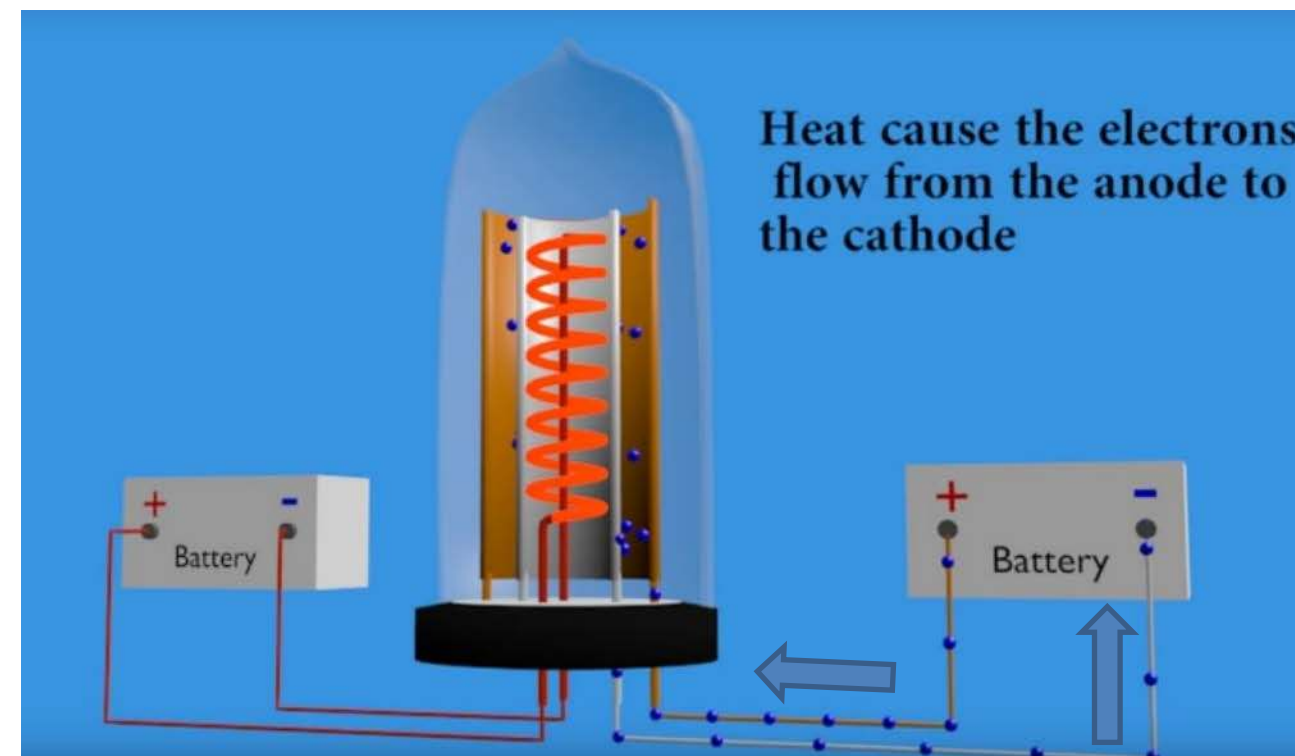
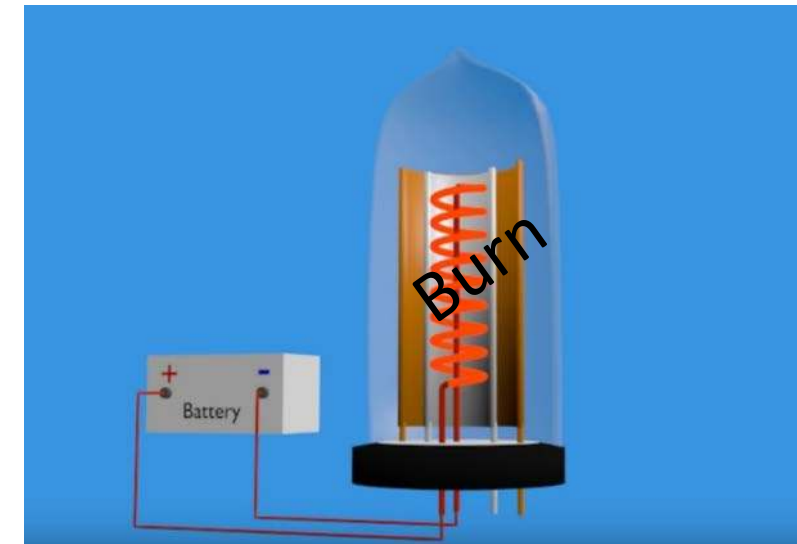
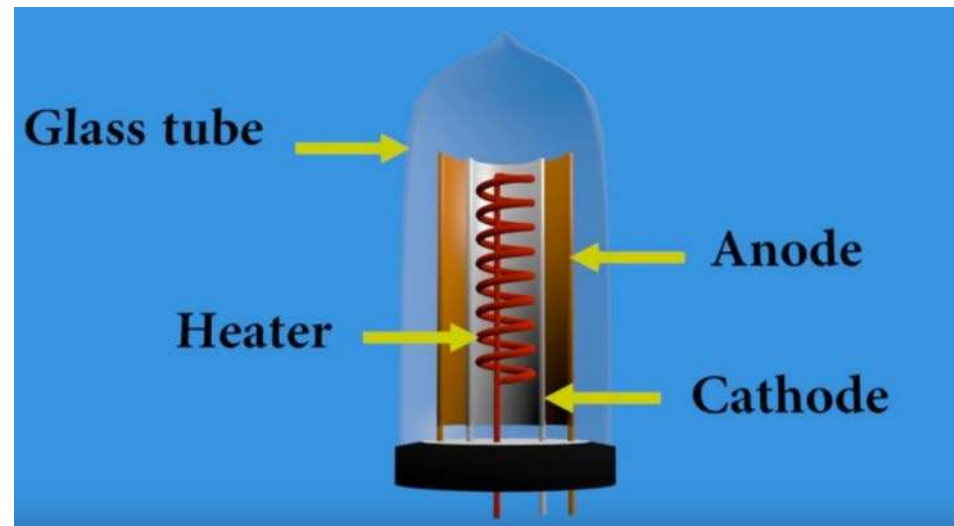


<https://youtu.be/aDN4s8ElxqE?t=144>

1.1.2 Vacuum Tube

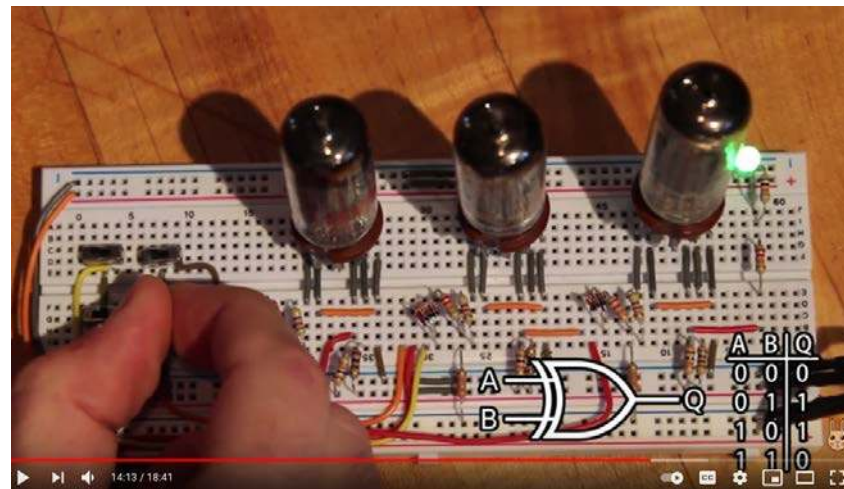
- Vacuum Tube was developed from *Incandescent lamp*.
- When a *tungsten⁷⁴* lead gets heat, it spreads electron by moving to an anode plate.
- A grid plate uses controls number of electron moving from the tungsten lead to an anode plate.





<https://www.youtube.com/watch?v=K6BgZ8s1Vuw>

Vacuum tube and future

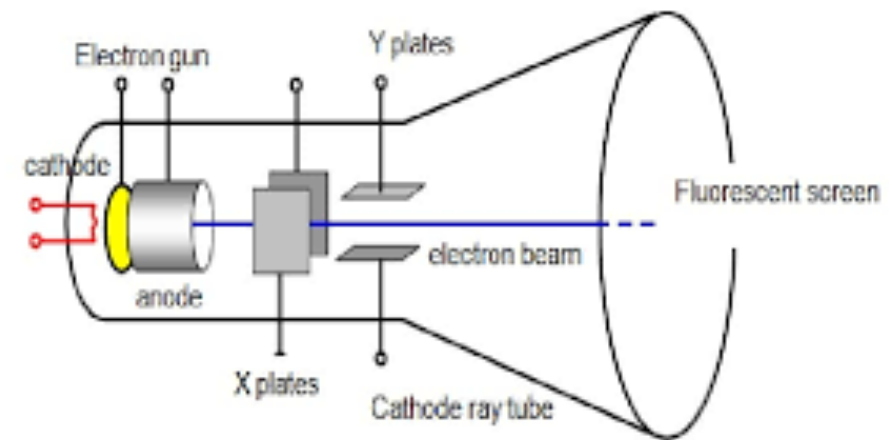
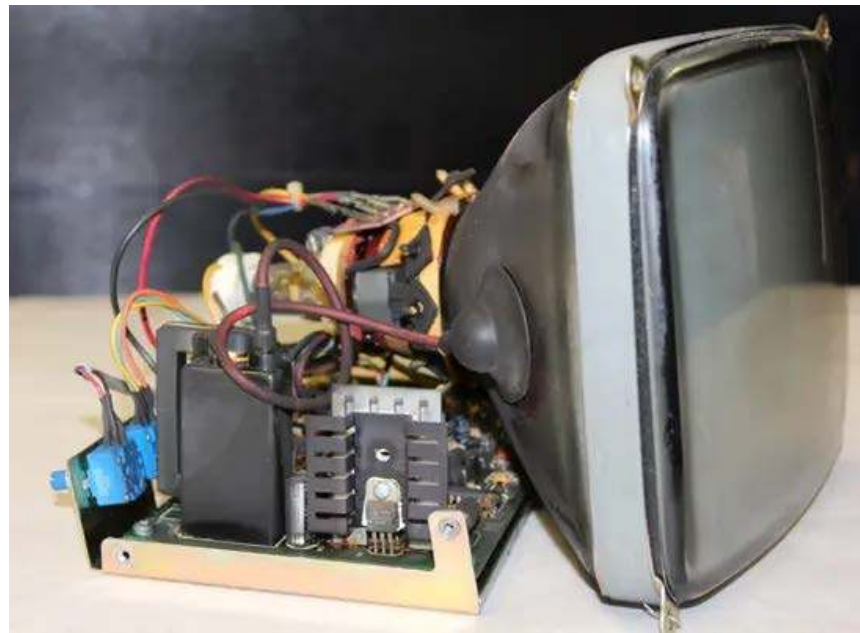


https://youtu.be/N-Sc6k_rITM?t=853

4.) Future: The Nano Vacuum Tube

Vacuum tubes may make a comeback and replace standard microchips. Engineers have been able to build a structure in phosphorus doped silicon and use nanotubes to build a switch. These devices can operate 10 times faster than silicon transistors.

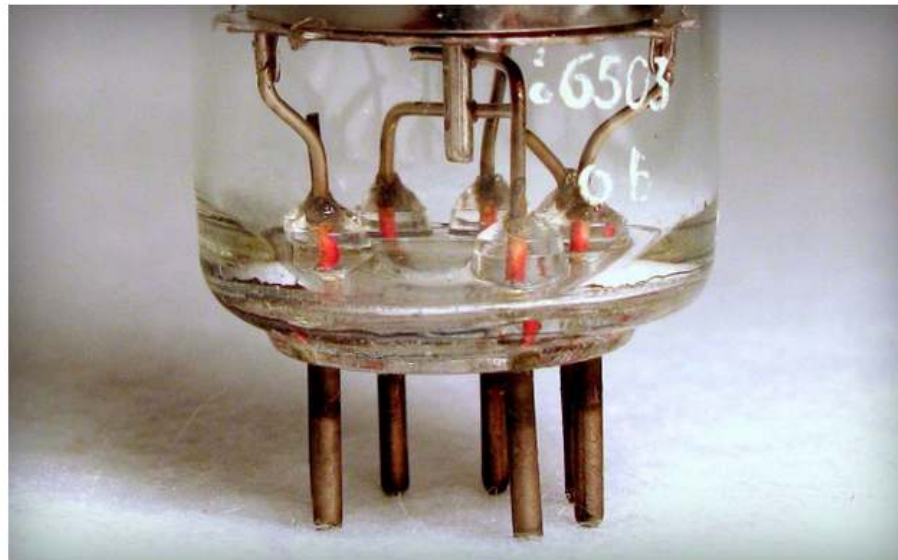
Cathod Ray Tube (CRT)



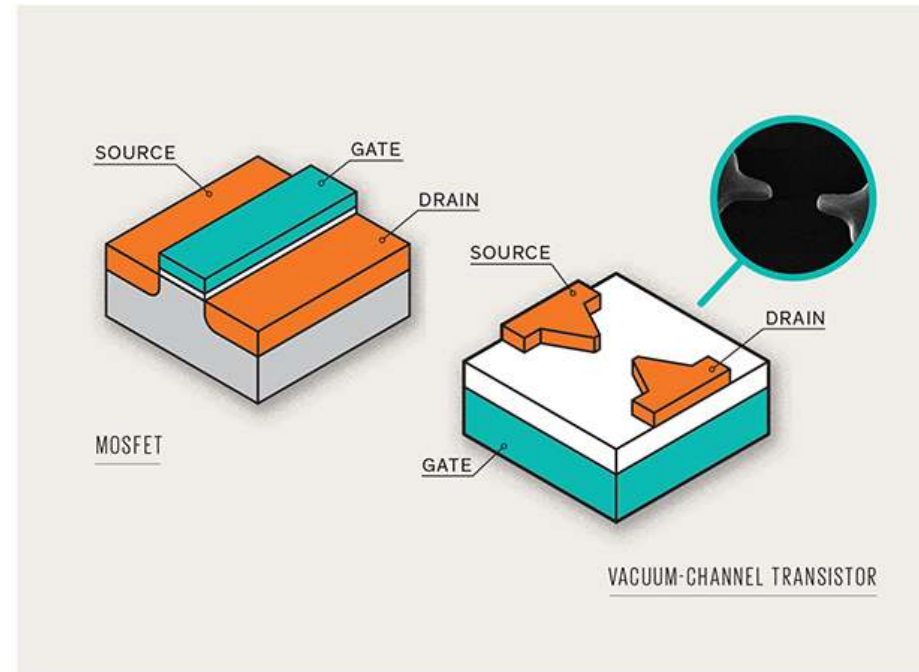
The vacuum tube strikes back: NASA's tiny 460GHz vacuum transistor that could one day replace silicon FETs

By Sebastian Anthony on June 24, 2014 at 8:12 am | [Comments](#)

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Way back in the salad days of digital computing (the 1940s and '50s), computers were made of vacuum tubes — big, hot, clunky devices that, when you got right down to it, were essentially glorified light bulbs. This is why early computers like the ENIAC weighed more than 27 tons and consumed more power than a small town. Later, obviously, vacuum tubes would be replaced by probably the greatest invention of all time — the solid-state transistor — which would allow for the creation of smaller, faster, cheaper, and more reliable computers. Fast forward to 2014, though, and the humble CMOS field-effect transistor (FET) is starting to show its age. We've pretty much hit the limit on shrinking silicon transistors any further, and they can't operate at speeds much faster than a few gigahertz. Which is why NASA's Ames Research Center is going back to the future with its new *vacuum transistor* — a nanometer-scale vacuum tube that, in early testing, has reached speeds of up to **460GHz**.

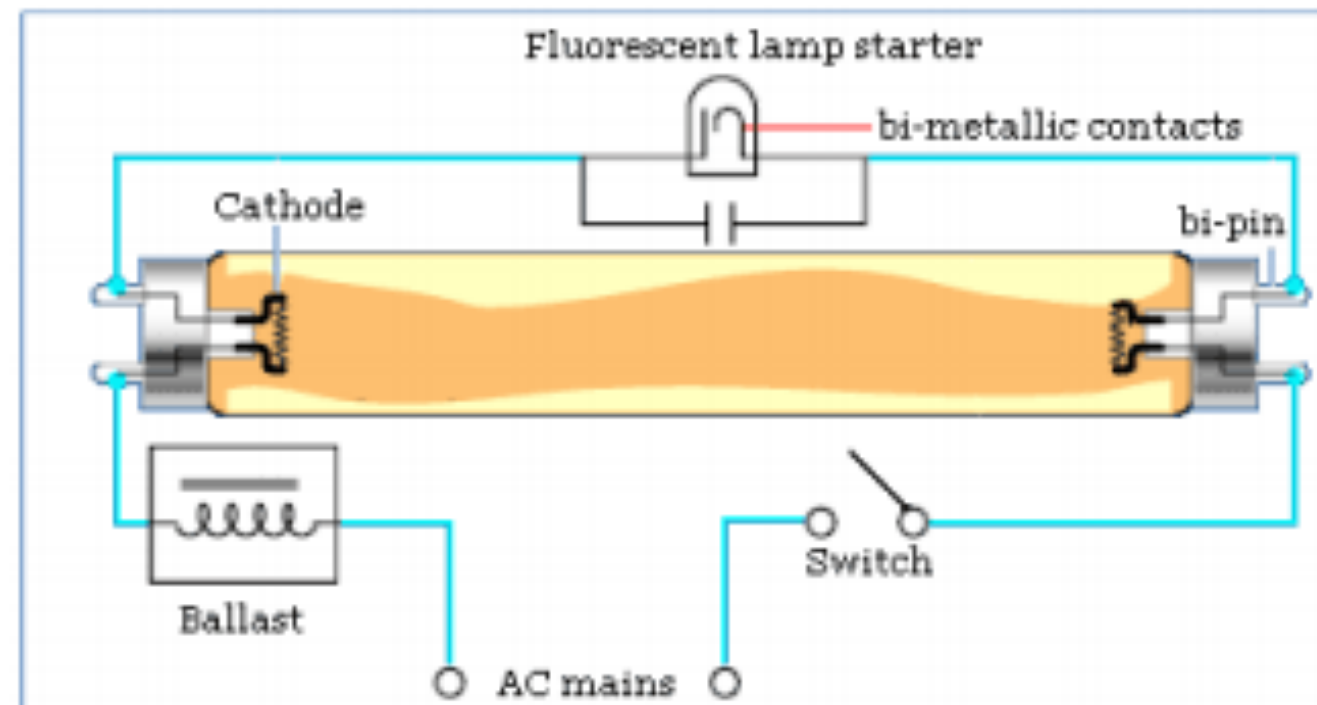


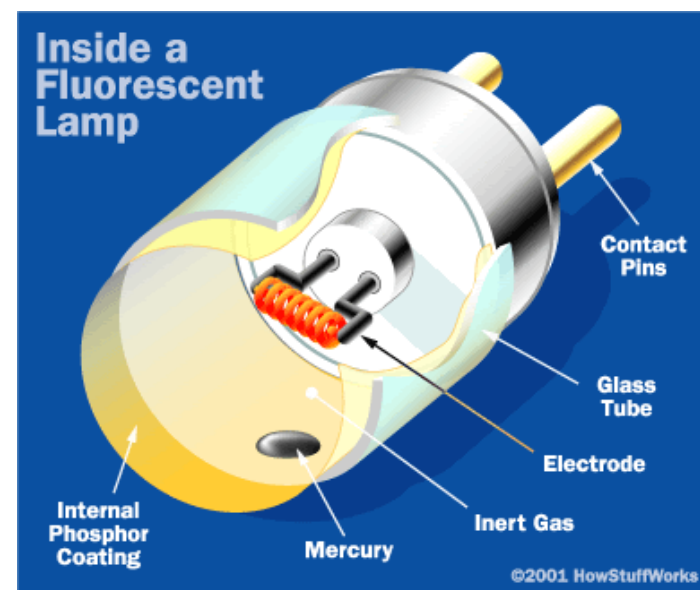
Standard MOSFET vs. vacuum-channel transistor [Image credit: IEEE Spectrum]

NASA's Ames Research Center has been working on-and-off for many years on the **vacuum-channel transistor**, which is essentially a vacuum tube that can be fabricated using conventional CMOS techniques. Instead of a gate sitting between the source and drain, there is... nothing. A vacuum. By a method known as *field emission*, electrons are drawn across the vacuum from the source to the drain when a current is applied to the gate (see diagram). By using field emission rather than the thermionic (hot) electron emission, these vacuum-channel transistors don't require a heat source. Importantly, they also don't require a vacuum — instead they use helium (it's sparse enough that the electrons have almost no chance of bumping into helium atoms while they traverse the few-nanometer gap between source and drain). The electrons also traverse the air gap a *lot* faster than if they had to pass through the gate electrode.

Activity1.2

- The fluorescence lamp has two small tungsten coils. Do you know why does the fluorescence lamp generates light? (Discussion)





1.1.3 Silicon Transistor

- Where are Silicon, Germanium, Boron, and Antimony on the periodic table?

The periodic table is color-coded to show different categories of elements. The legend indicates the following categories:

- State of matter (color of name):** GAS (red), LIQUID (blue), SOLID (green), UNKNOWN (grey).
- Subcategory in the metal-metalloid-nonmetal trend (color of background):**
 - Alkali metals (red)
 - Alkaline earth metals (orange)
 - Transition metals (blue)
 - Lanthanides (light blue)
 - Actinides (dark blue)
 - Post-transition metals (light green)
 - Metalloids (yellow)
 - Reactive nonmetals (dark green)
 - Noble gases (pink)
 - Unknown chemical properties (grey)

The elements are arranged in groups (columns) and periods (rows). The groups are labeled with Roman numerals and letters (e.g., IA, IIA, IIIA, etc.). The elements are color-coded according to the legend.

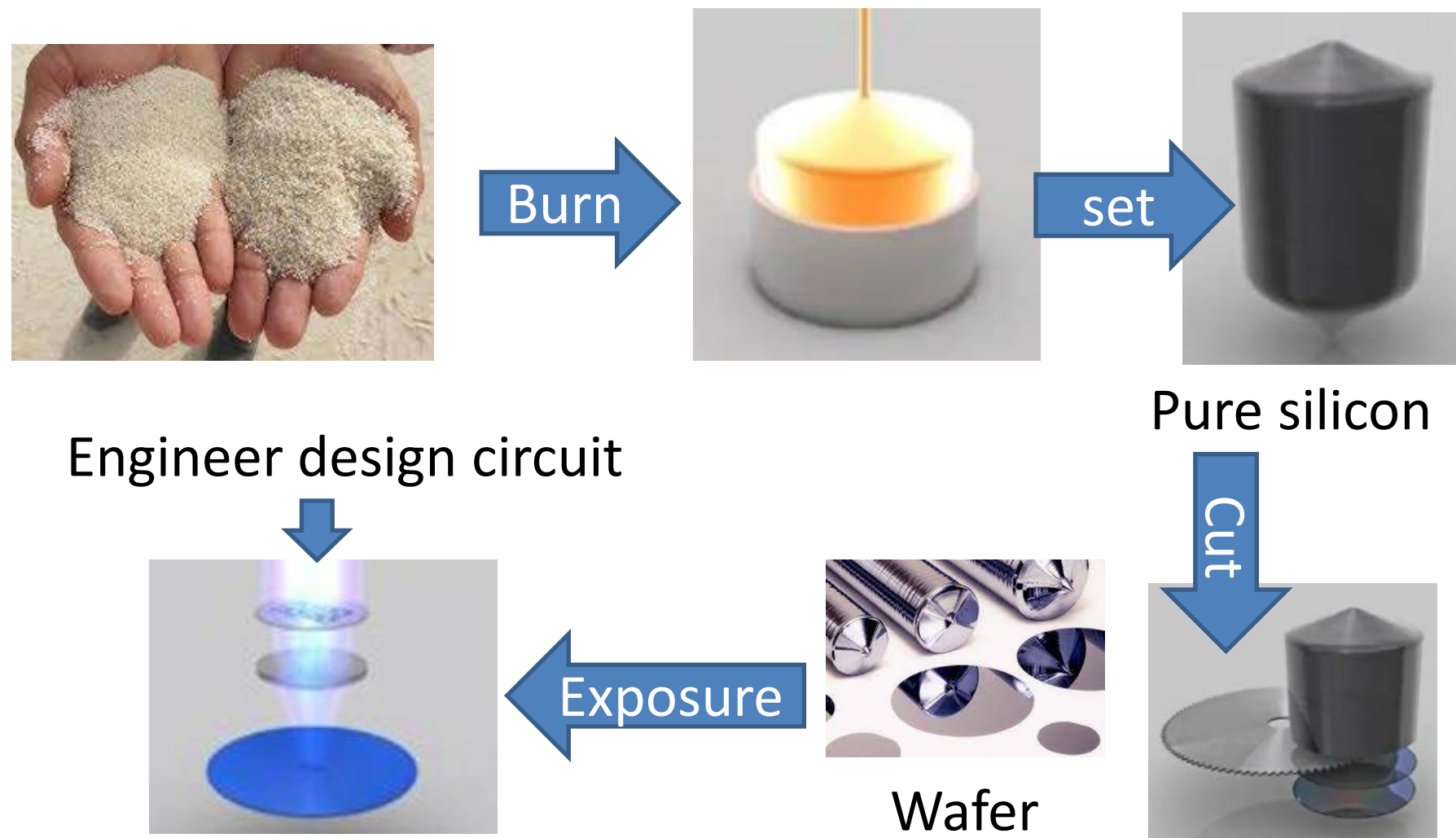
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
IA	IIA	IIIB	IVB	VB	VIB	VII	VIII	VIII	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 H Hydrogen 1.008	2 He Helium 4.003																
3 Li Lithium 6.94	4 Be Beryllium 9.012											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 18.99	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31											13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 51.99	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38	31 Ga Gallium 69.72	32 Ge Germanium 72.64	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium 98.91	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.91	54 Xe Xenon 131.29
55 Cs Cesium 132.91	56 Ba Barium 137.33	57-71 Lanthanides	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium 209	85 At Astatine 210	86 Rn Radon 222
87 Fr Francium 223	88 Ra Radium 226	89-103 Actinides	104 Rf Rutherfordium 261	105 Db Dubnium 262	106 Sg Seaborgium 266	107 Bh Bohrium 264	108 Hs Hassium 277	109 Mt Meitnerium 268	110 Ds Darmstadtium 271	111 Rg Roentgenium 272	112 Cn Copernicium 285	113 Nh Nihonium 284	114 Fl Flerovium 289	115 Mc Moscovium 288	116 Lv Livermorium 293	117 Ts Tennessine 294	118 Og Oganesson 294
57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium 145	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97			
89 Ac Actinium 227	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium 237.05	94 Pu Plutonium 244.06	95 Am Americium 243.06	96 Cm Curium 247.07	97 Bk Berkelium 247.07	98 Cf Californium 251.08	99 Es Einsteinium 252.08	100 Fm Fermium 257.10	101 Md Mendelevium 258.11	102 No Nobelium 259.10	103 Lr Lawrencium 262.11			

1.1.3 Silicon chip-making process

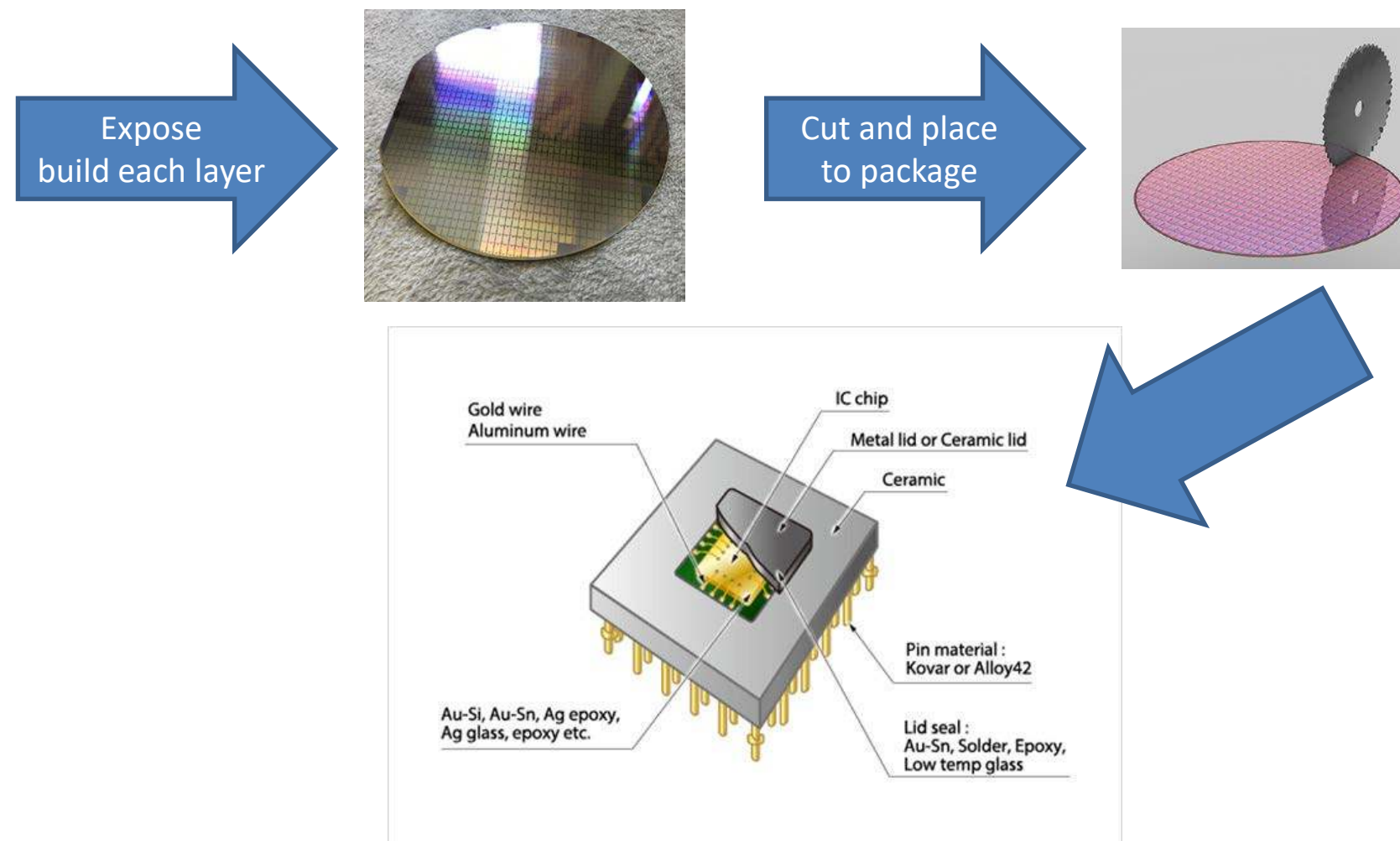
- Silicon
 - Make from the ordinary sand, quartz, rock crystal, amethyst, agate, flint, jasper, and opal.



1.1.3 Silicon chip-making process

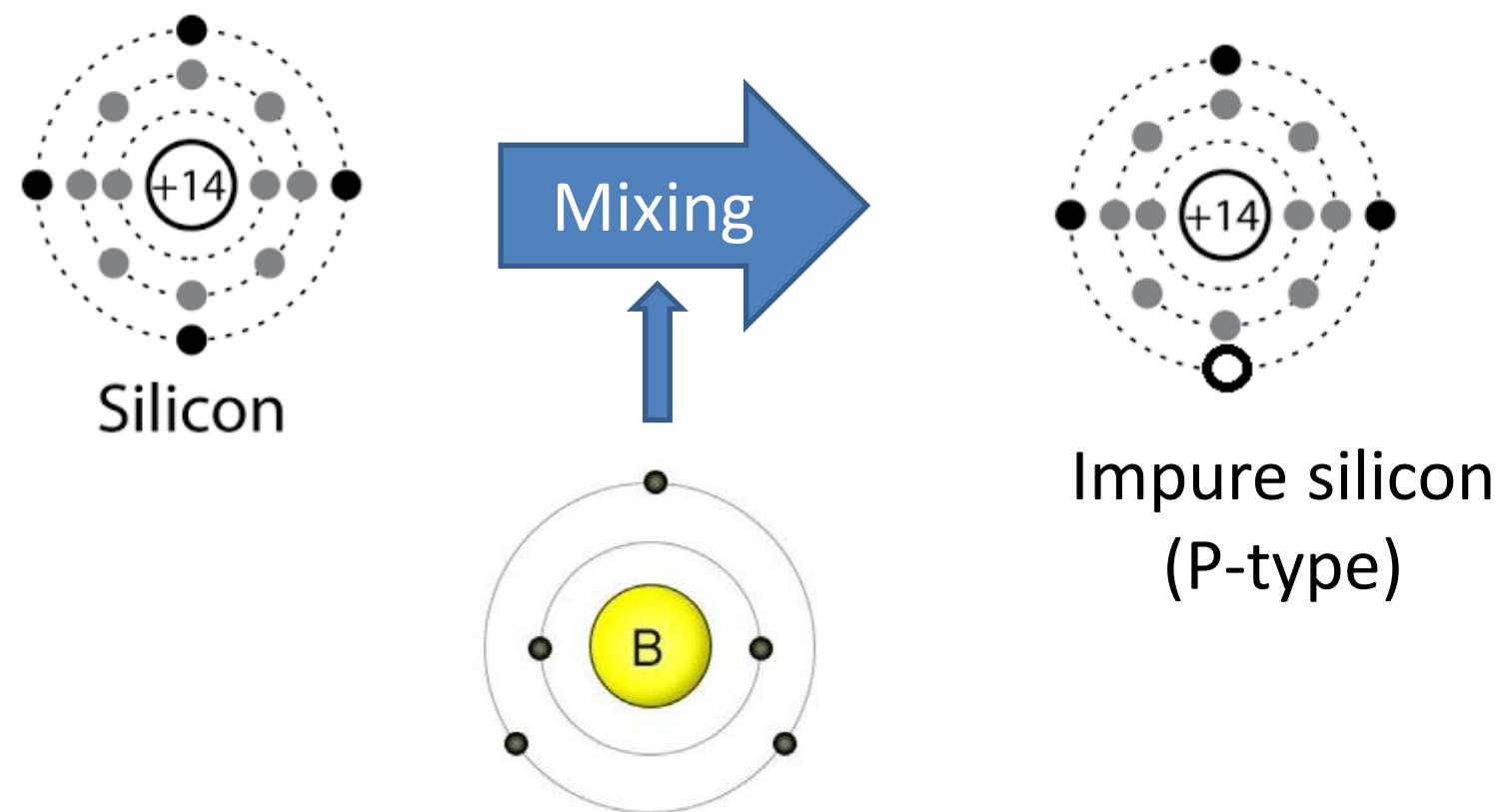


1.1.3 Silicon chip-making process



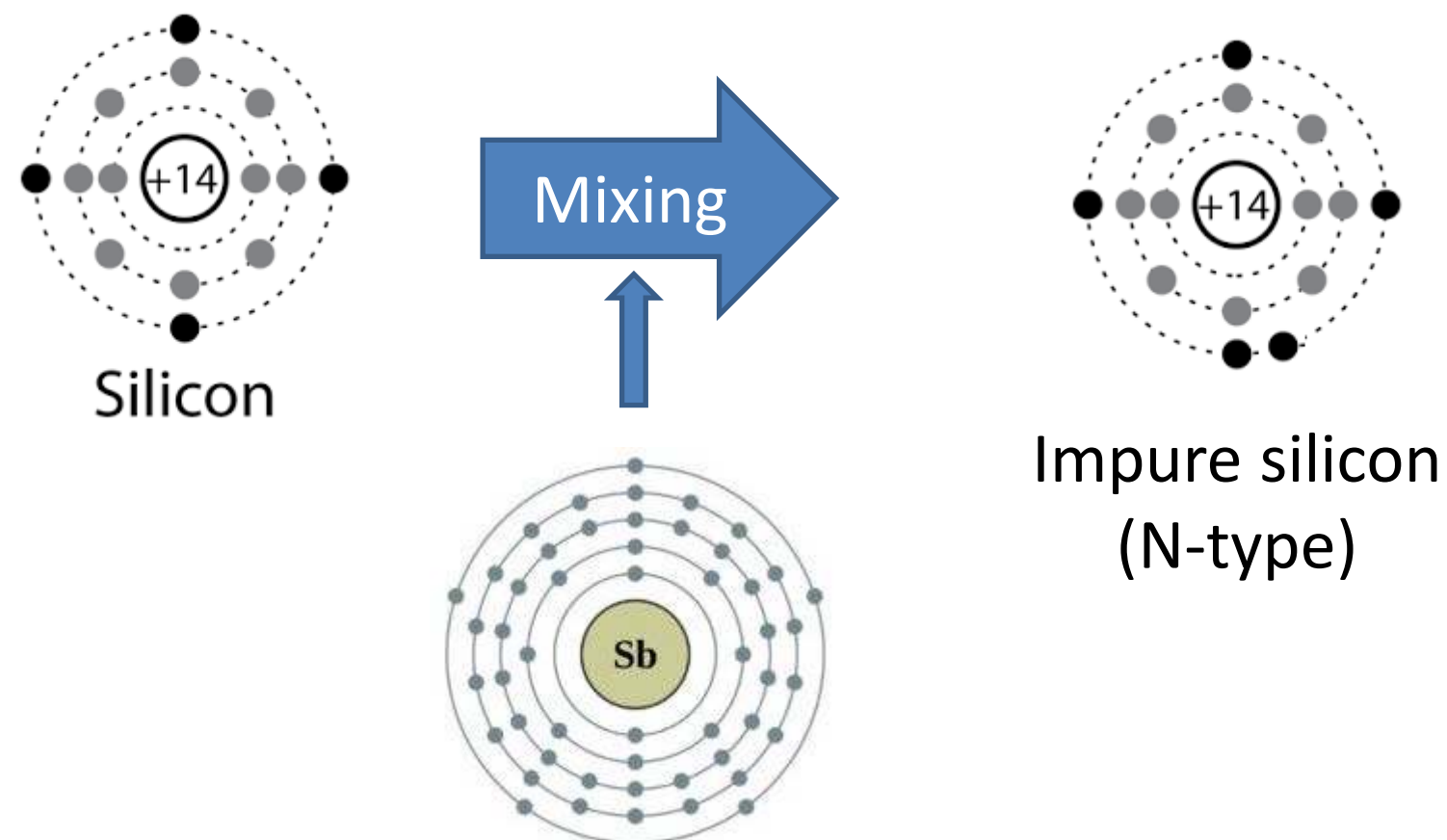
The main point of silicon chip

- Change from pure silicon to impure silicon by mixing boron to be P-type

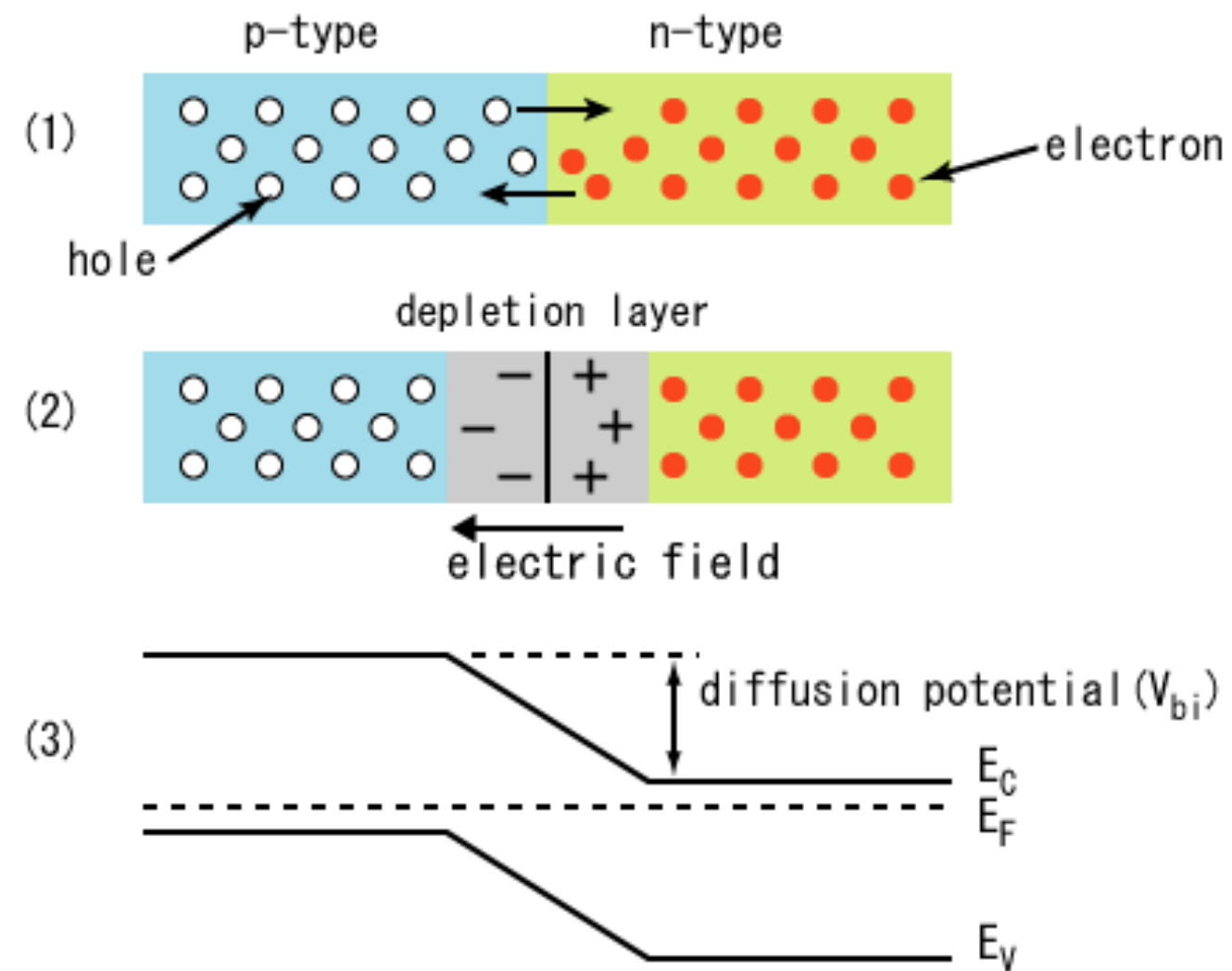


The main point of silicon chip

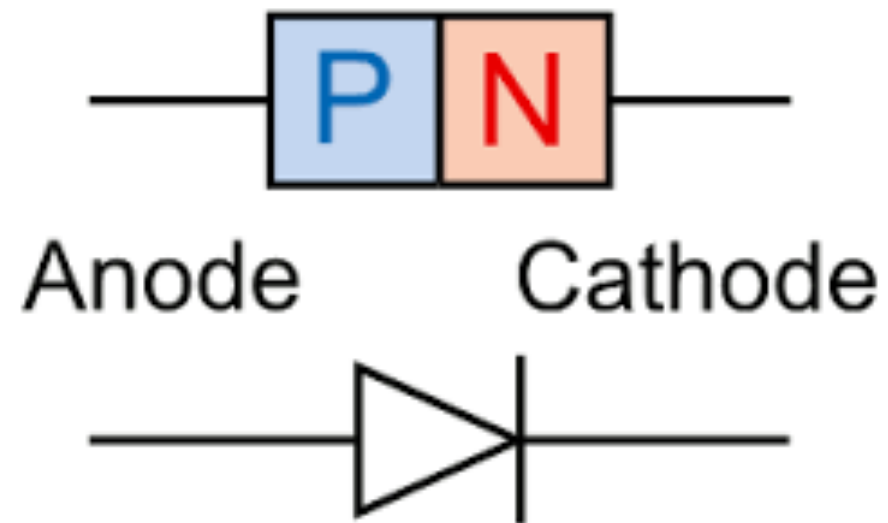
- Change from pure silicon to impure silicon by mixing antimony แร่พวง to be N-type



N-type and P-type

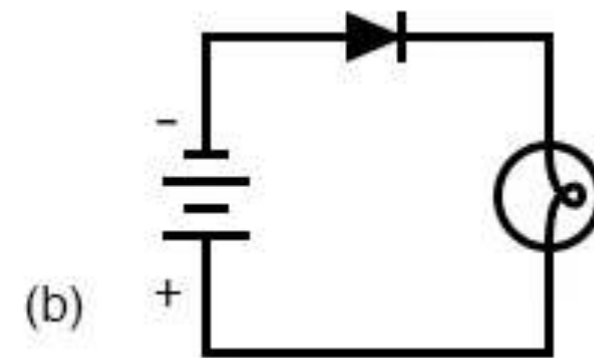
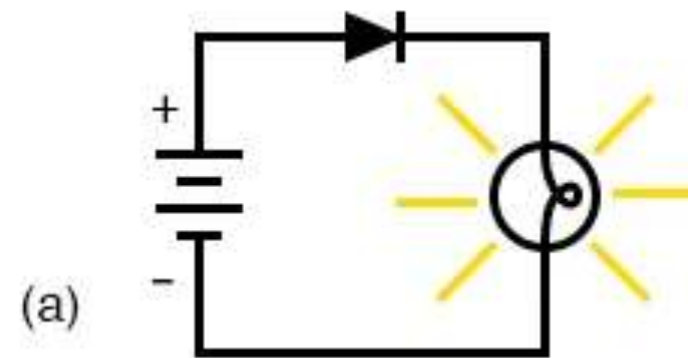


Diode

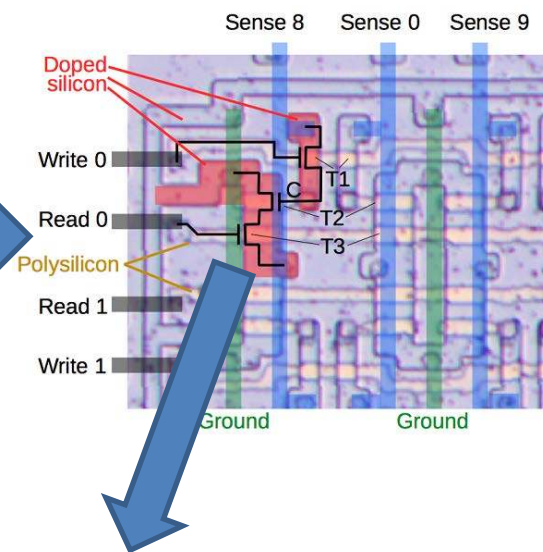
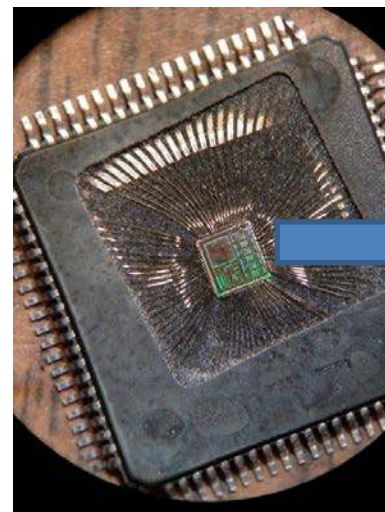


Activity 1.3

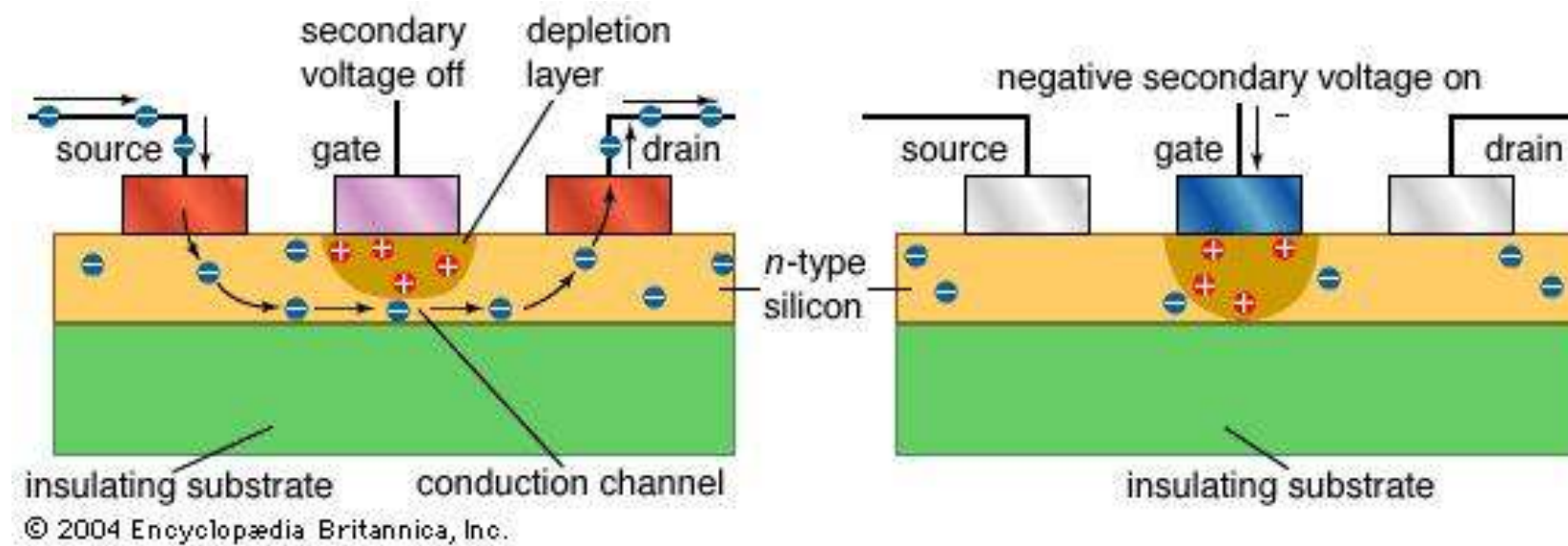
- Let's test a diode on Thinkercad.



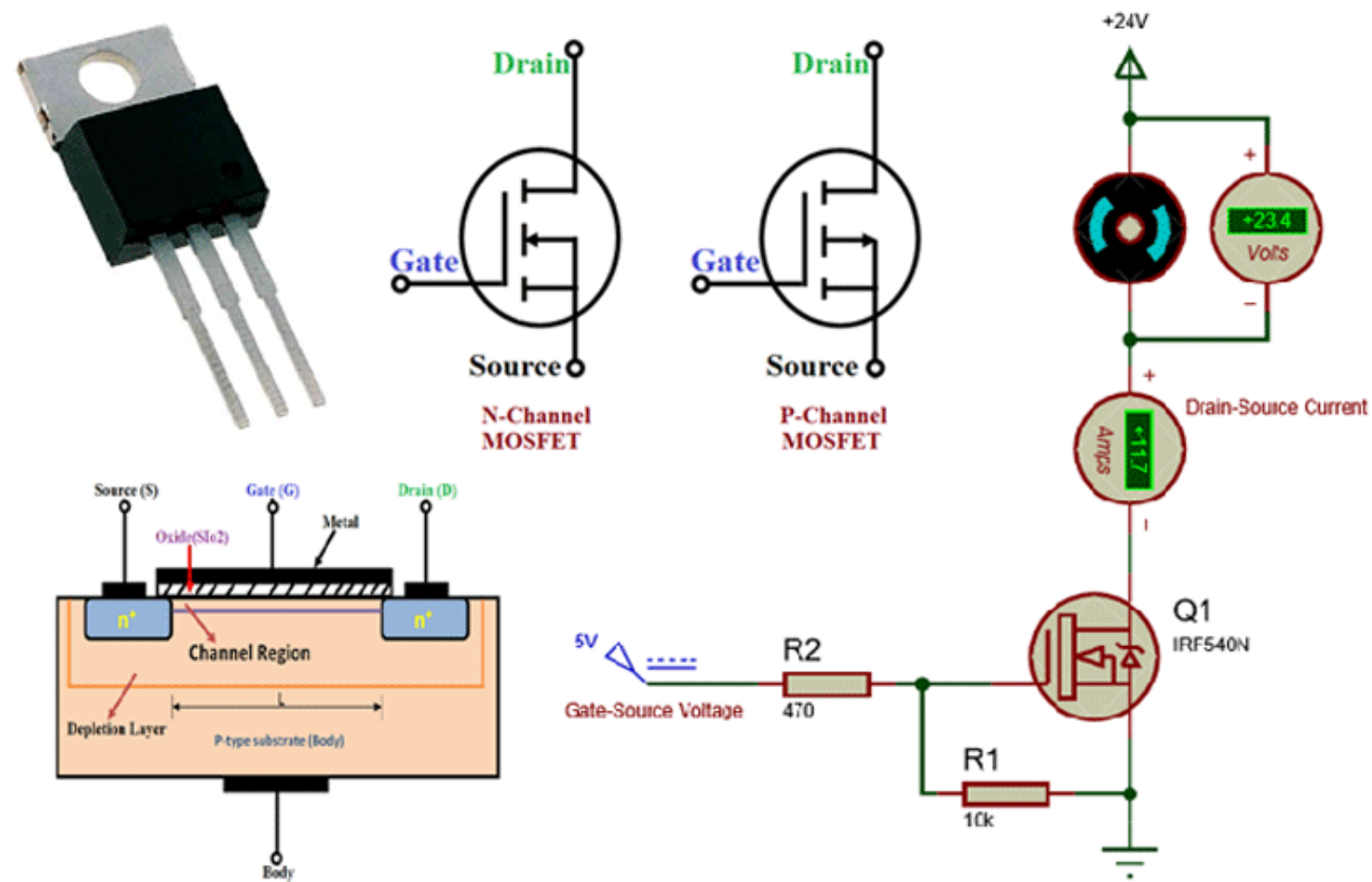
1.1.3 Silicon Transistor



- The **gate** pin controls number of the electron move from the **source** pin to the **drain**.

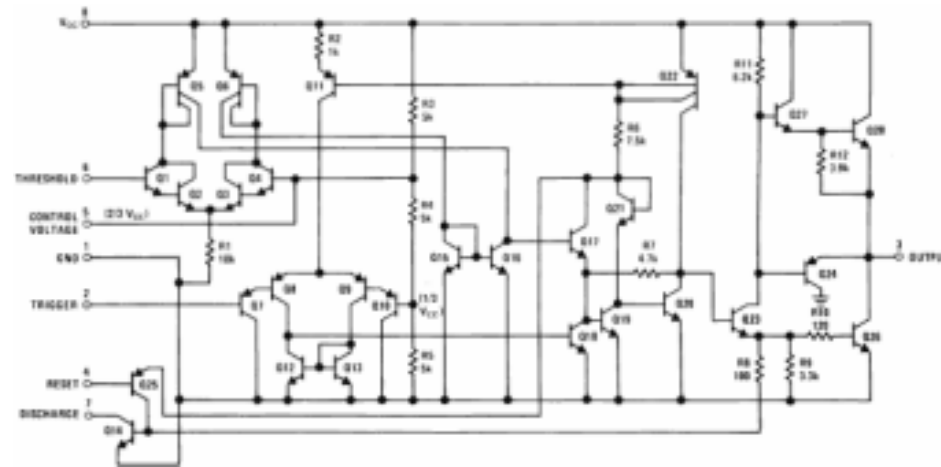


FET and Mosfet has Drain Gate Source



1.1.4 Integrated Circuit

- Discrete Circuit
- Integrated Circuit



1.1.4 Integrated Circuit: Chip Scale

Name	Number of Transistors
Small Scale IC (SSI)	10
Medium SI (MSI)	10 – 1000
Large SI (LSI)	1K – 100K
Very LSI (VLSI)	100K – 1M
Ultra LSI (ULS)	> 1M
(Nvidia GA100)	>54Bilion Transistor

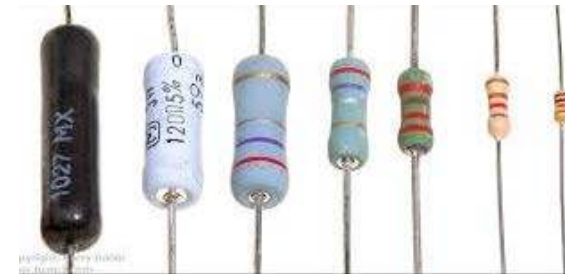
Number of transistor
relate to the circuit
complexity

Year	Size of the transistor in chip
1971 - 1985	10 μm – 1 μm
1989 - 1999	800nm – 180nm
2001 - 2010	130nm – 32nm
2012 - 2017	22nm – 10nm
2018	7nm
2020	5nm
Appr. 2022	3nm

Transistor size relate to
power consumption and
switching speed

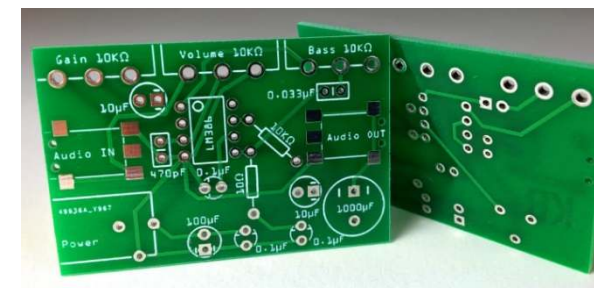
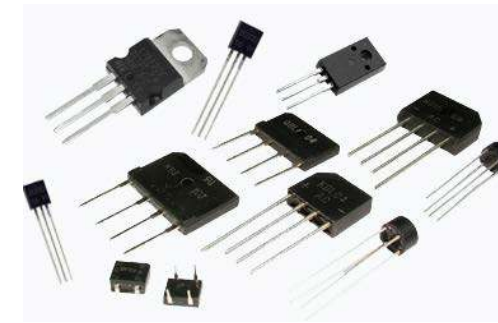
Electronic devices

- Passive device
 - Capacitor
 - Resistor
 - Inductor
 - Transformer



Electronic devices

- Active device
 - Diode
 - Transistor
 - Light Emitting Diode (LED)
 - IC, Chip
- Component
 - Print Circuit Board (PCB)



Activity 1.4

- Measure resistance and capacitance of R and C with an analog multimeter.
- Measure voltage from various battery size with an analog multimeter

1.2 The inside of Computer

- Processor
- Main memory
- System bus
- I/O Module

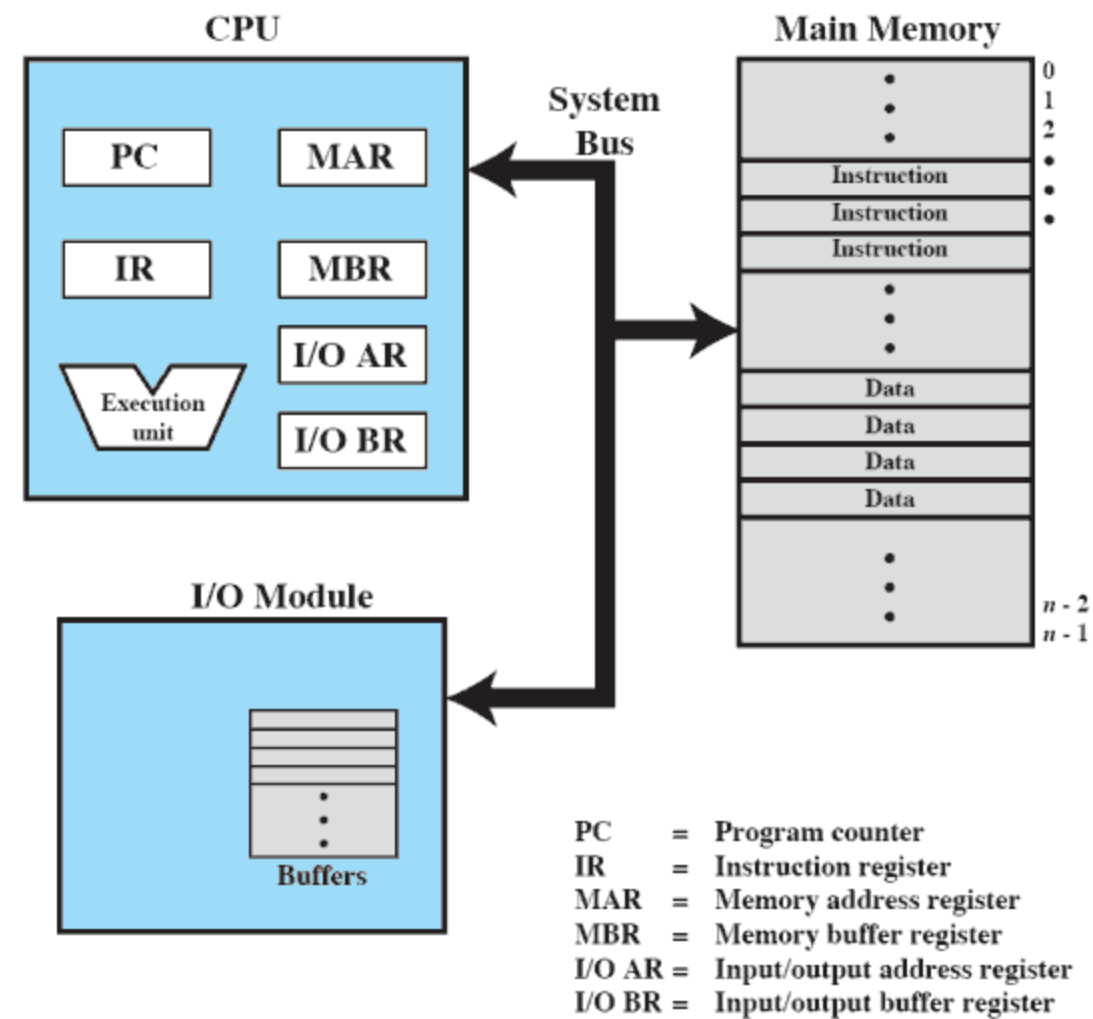
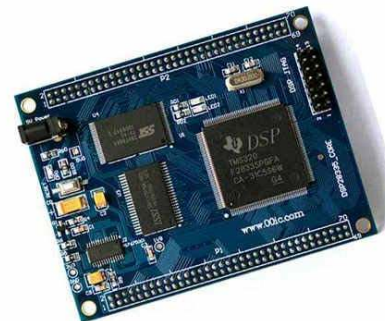
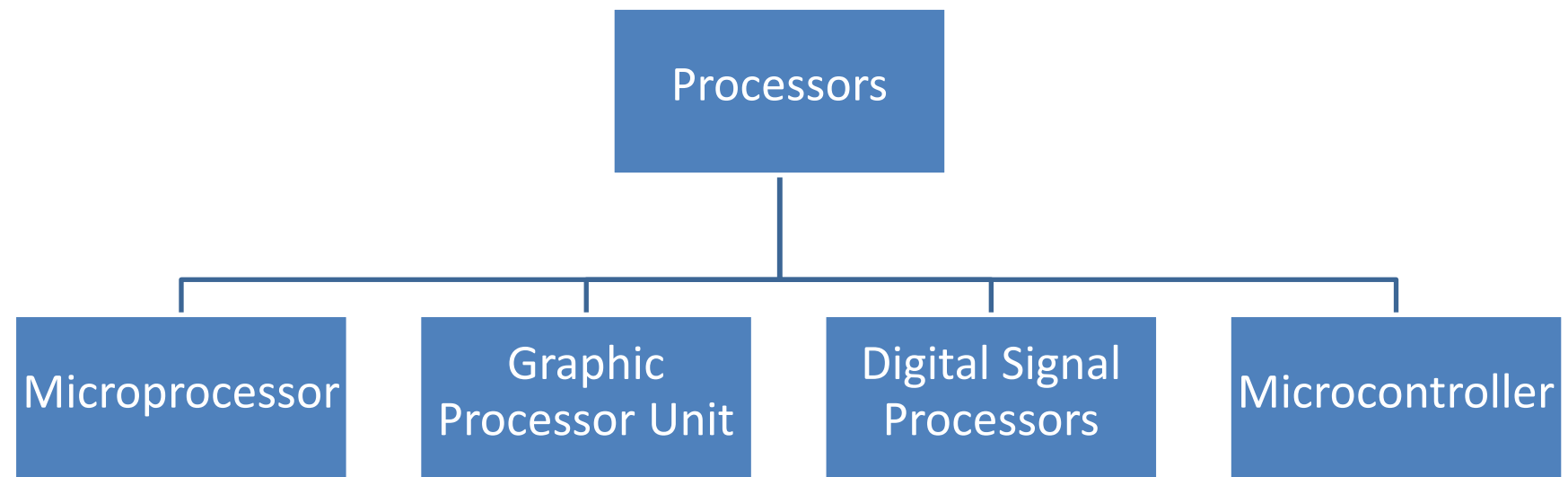
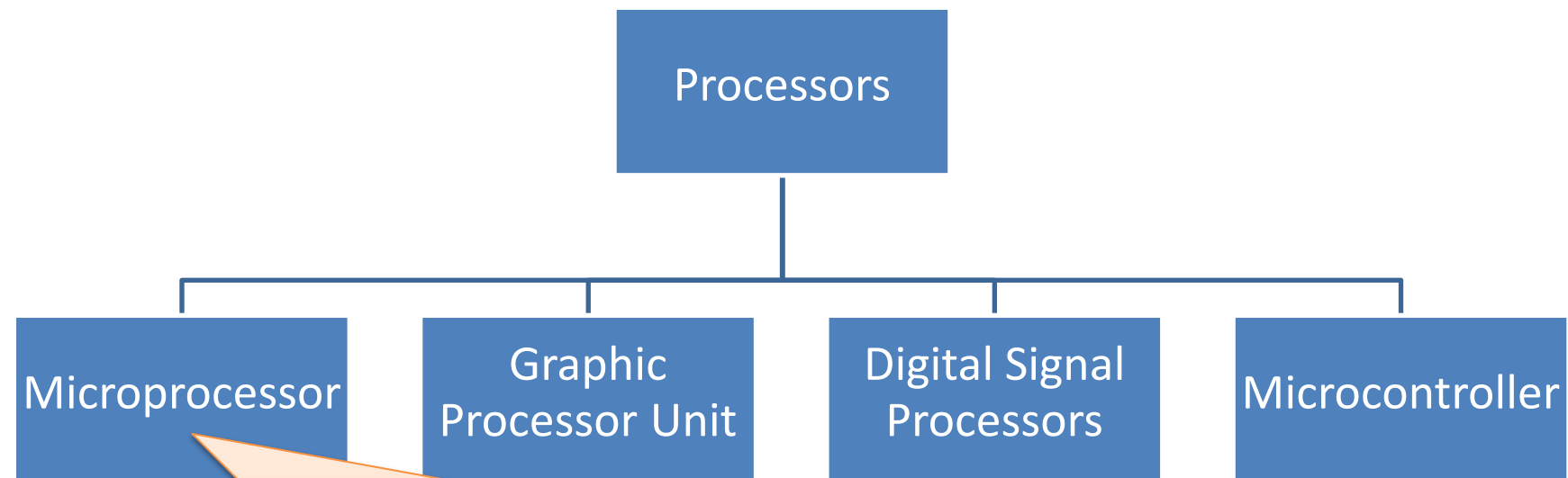


Figure 1.1 Computer Components: Top-Level View

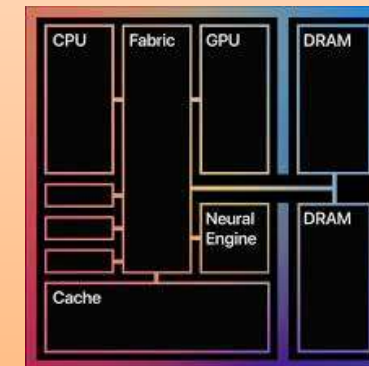
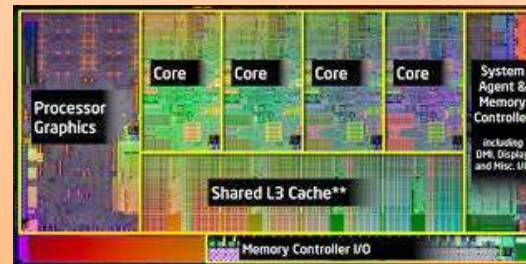




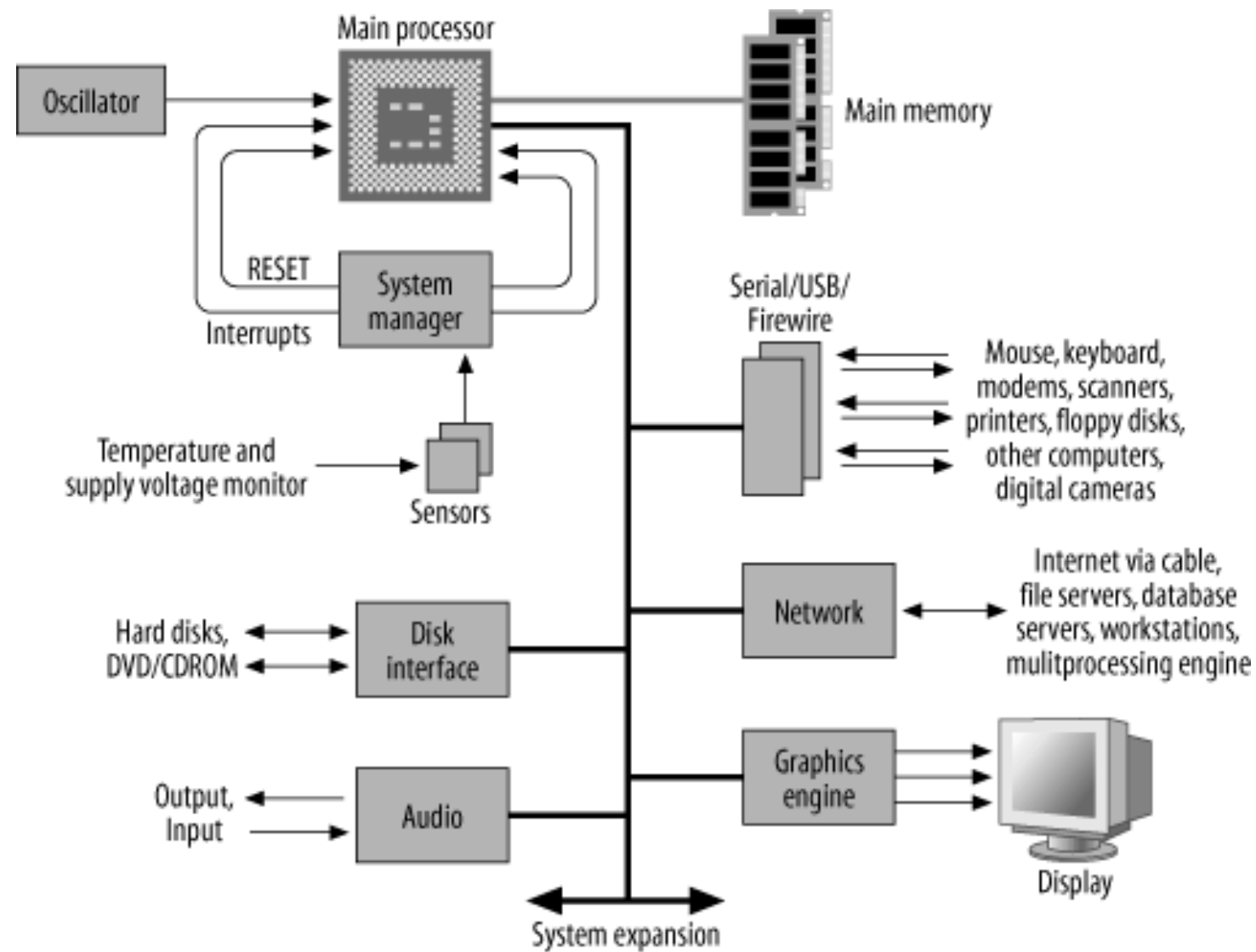
- Design instruction set for the general work
- Parallel processing
- No Input output ports (I/O)
- Some microprocessors are integrated GPU.
- Advance microprocessors are integrated GPU, I/O, and Memory by called **System-on-chip (SOC)**

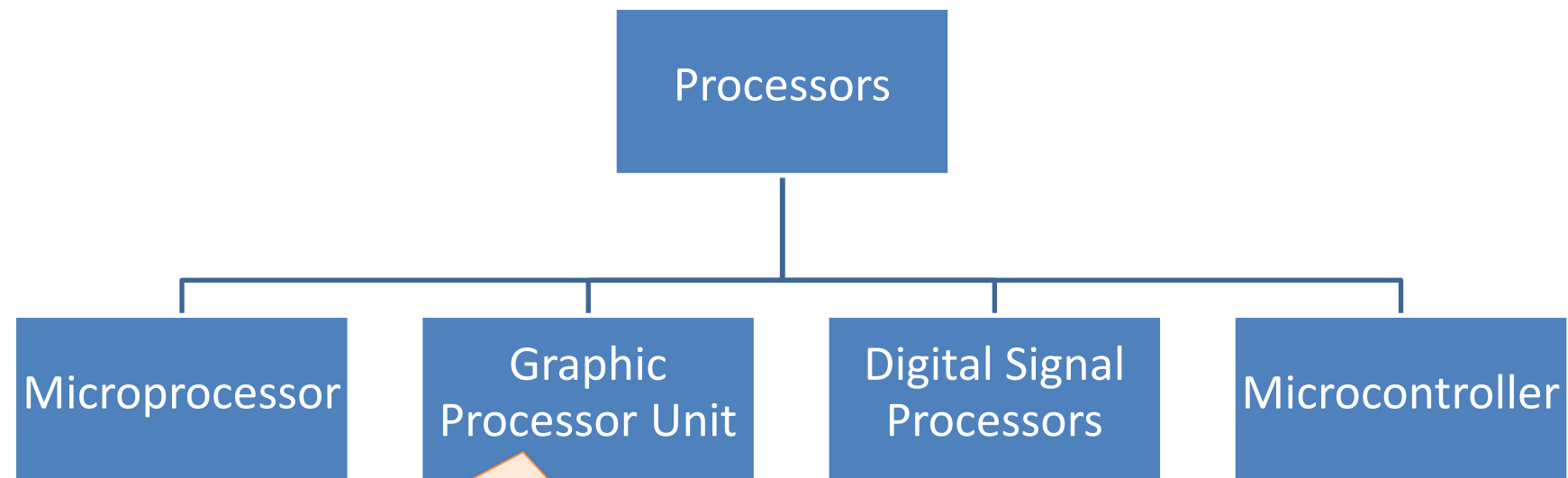
- Example

- Intel, Core-i7
- AMD, Ryzen
-

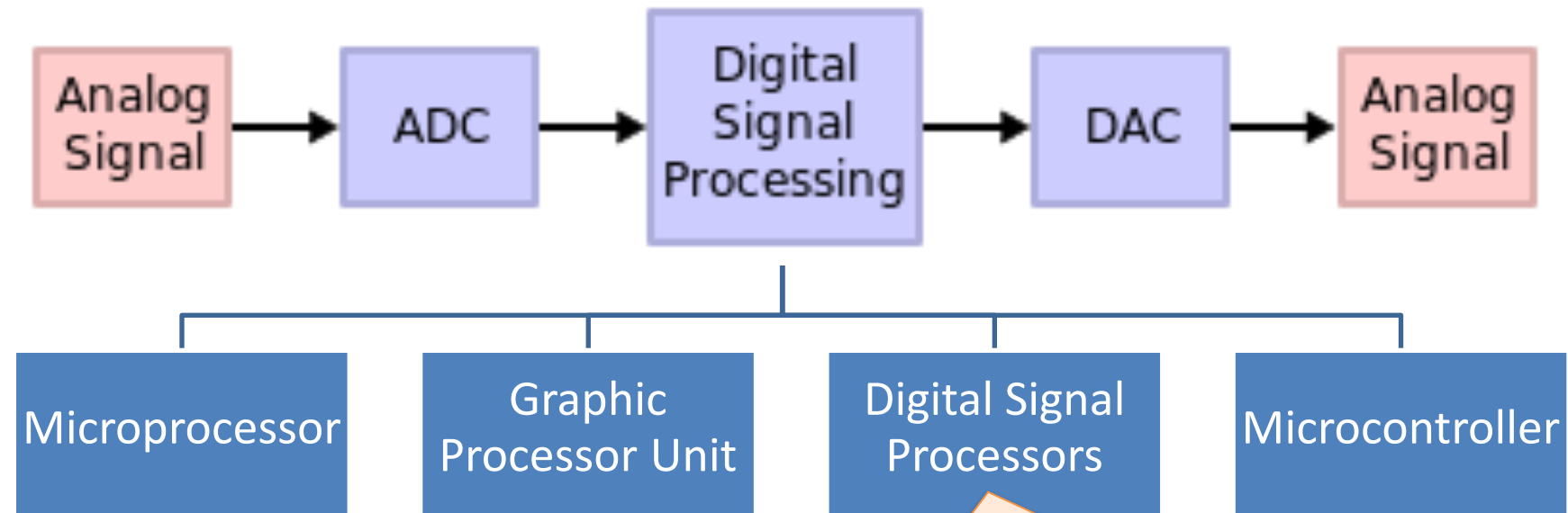


Computer systems

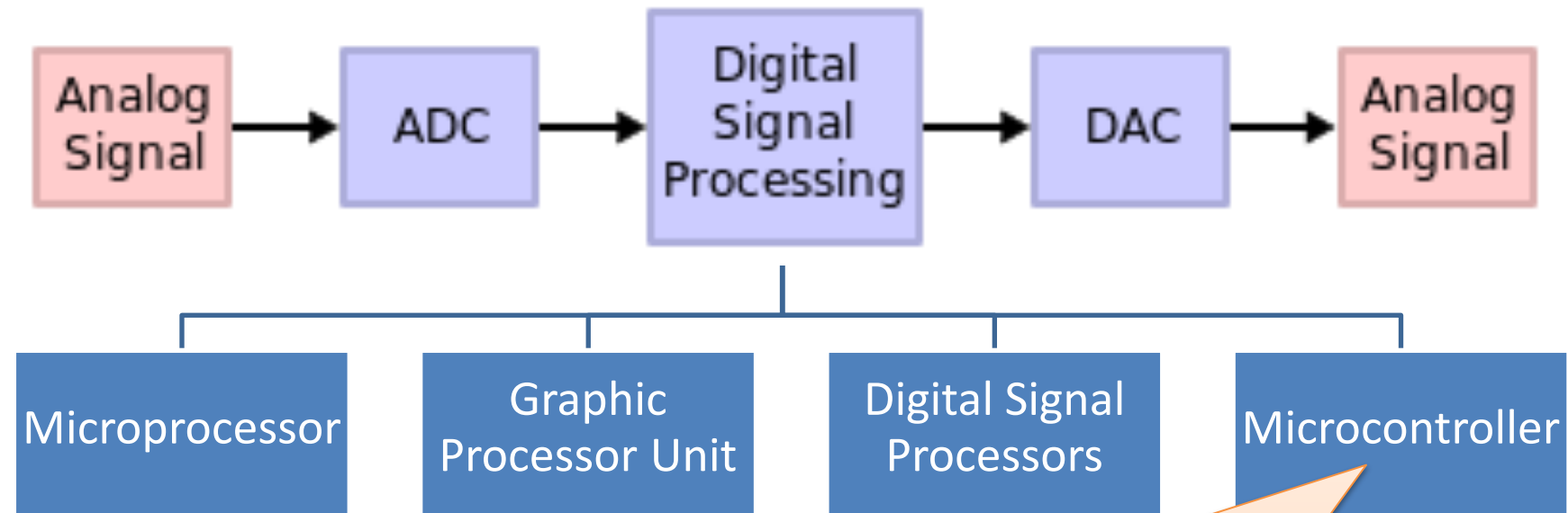




- Specific instruction set for vector and graphic computing and video rendering
- Dual ports memory interface
- Multicores
- Example family chips
 - GeForce, Radeon
 - Quadro, FirePro

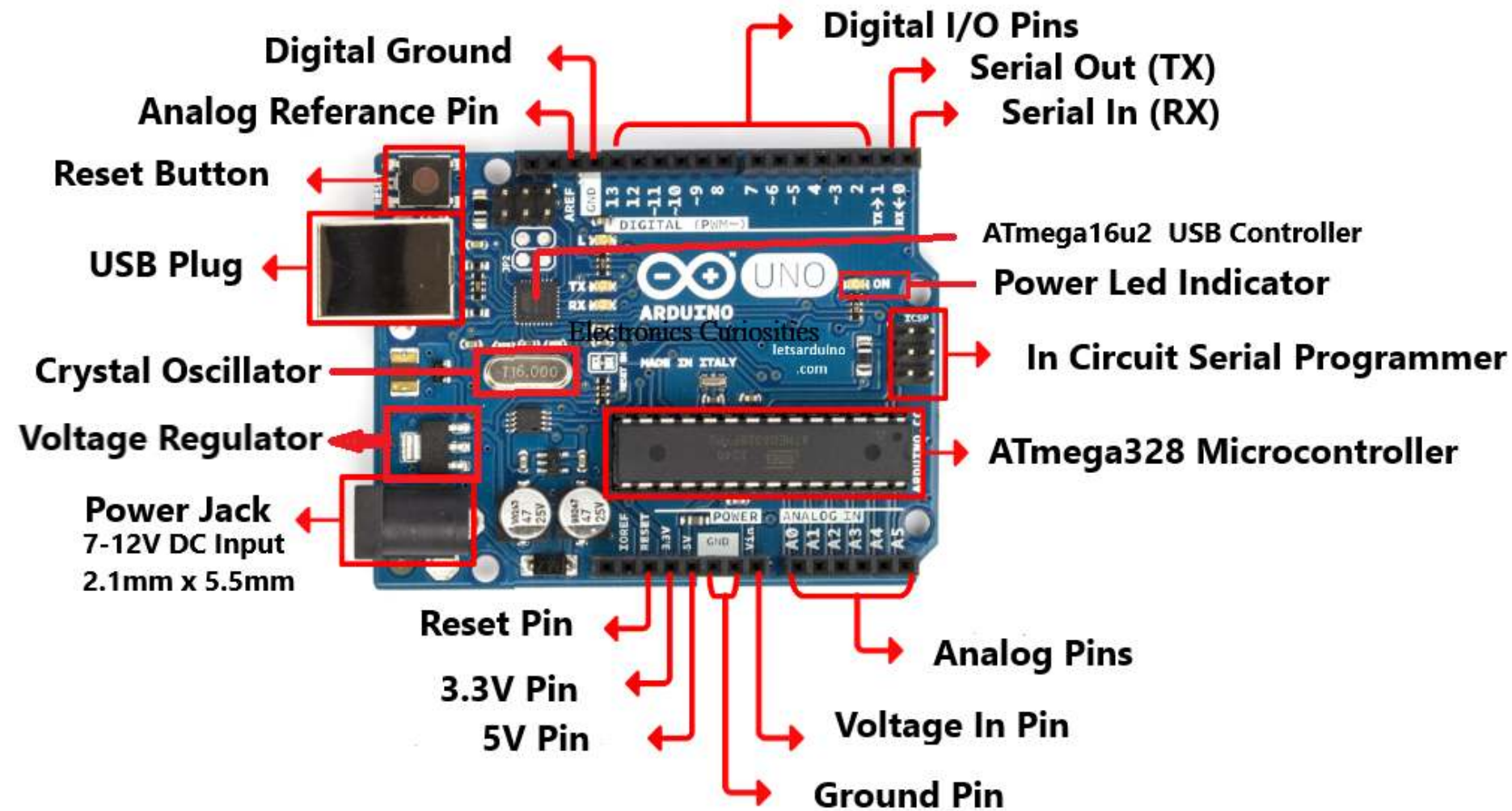


- Specific instruction set for streaming signal
 - Encoding audio/video
 - Decoding audio/video
- Example family chips
 - C6000 (Texas Instrument)
 - SHARC (Analog Devices)
 - EMU10K (Creative)



- Specific instruction set for input/output (I/O) controlling
- Design to control appliance machines such as washing machines, microwaves, toys, etc.
- Integrated I/O ports, timers, memory, Wifi, analog to digital and digital to analog.
- Example family chips
 - PIC
 - MCS-51
 - ARM

Arduino board



Activity1.5

- Identify the type of chip on the list below. You can search from Internet.
 - Z80
 - 8051
 - 68HC11
 - PIC 16Fxx
 - Atmega 328p
 - 8086
 - Geforce 1060
 - TMS32010
 - MAS3507